

The Unrestricted Even-Input Gaussian-Saturation Conjecture Is False, and Its Log-Concave Repair Is Known

Literature-status packet for arXiv:2401.00427

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Status: literature already answered. The universal all-even Gaussian-saturation statement proposed as equation (3.5) of the source is false. A later theorem proves the statement under log-concavity, and a subsequent paper explains why log-concavity cannot be omitted.

1 The source target

For inverse Brascamp–Lieb data (L, c, Q) , the source defines

$$\Lambda(L, c, Q; \mathbf{f}) = \int_{\mathbb{R}^d} e^{-\pi \langle x, Qx \rangle} \prod_{i=1}^m f_i(L_i x)^{c_i} dx.$$

In Section 3.2, PDF page 15, it proposes equation (3.5): for every datum, the infimum over normalized nonnegative even inputs should equal the infimum over centered Gaussian inputs. The statement is offered as a degenerate extension of Barthe–Wolff Gaussian saturation and as a route to the Kolesnikov–Werner many-function Santaló conjecture.

2 What the later literature proves

Nakamura and Tsuji subsequently proved a sharp positive theorem. Their Theorem 1.3 in arXiv:2409.13611 states that for positive exponents and an arbitrary symmetric quadratic kernel, centered Gaussians exhaust the inverse Brascamp–Lieb lower bound when all inputs are nonnegative, even, integrable, and *log-concave*. Their Theorem 4.1 also establishes the linked Kolesnikov–Werner exponential-kernel inequality.

Milman then showed in Section 1.1 of arXiv:2501.11018 that log-concavity is essential. If the same lower bound held for all even inputs, his reduction would imply

$$\mathbb{P}(X_1 \in K, X_2 \in L) \geq \mathbb{P}(X_1 \in K)\mathbb{P}(X_2 \in L) \tag{1}$$

for every centered correlated Gaussian pair and every origin-symmetric measurable K, L . Applying (1) both to L and to its complement forces equality, hence independence of those two events, which is false for a generic correlated pair.

Milman records the resulting inverse Brascamp–Lieb counterexample with $N = 2$, $m = 2$, $n_1 = n_2 = 1$, $c_1 = c_2 = 1$,

$$h_1 = \mathbf{1}_{[-1,1]}, \quad h_2 = \mathbf{1}_{\mathbb{R} \setminus [-1,1]},$$

and the quadratic data induced by a generic positive-definite covariance matrix Σ . Concretely one may take a standard Gaussian pair with correlation $1/2$: strict positive association of the two interval events makes the interval/complement version of (1) fail.

3 Disposition and scope

The source’s unrestricted “for any datum, all even inputs” conjecture is therefore answered negatively in the later literature. Its natural log-concave repair is answered affirmatively. This status identification does not claim that every broader version of the original Kolesnikov–Werner conjecture for arbitrary radial kernels follows from these papers; it resolves precisely the Gaussian-saturation route proposed in equation (3.5) and its cited exponential-kernel application.

References

- [1] S. Nakamura and H. Tsuji, *The functional volume product under heat flow*, arXiv:2401.00427v2, 2024.
- [2] S. Nakamura and H. Tsuji, *A generalized Legendre duality relation and Gaussian saturation*, arXiv:2409.13611, 2024; later published in *Inventiones Mathematicae*.
- [3] E. Milman, *Gaussian Correlation via Inverse Brascamp–Lieb*, arXiv:2501.11018v2, 2025; to appear in *Probability Theory and Related Fields*.